

torch.atanh#

<https://pytorch.org/docs/stable/generated/torch.atanh.html>

Description:

Returns a new tensor with the inverse hyperbolic tangent of the elements of input. The domain of the inverse hyperbolic tangent is $(-1, 1)$ and values outside this range will be mapped to NaN, except for the values 1 and -1 for which the output is mapped to $\pm\text{INF}$ respectively. input (Tensor) – the input tensor. out (Tensor, optional) – the output tensor.

Function Signature

```
torch.atanh(input: Tensor, *, out: Optional[Tensor]) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(4).uniform_(-1, 1)
>>> a
tensor([ -0.9385,  0.2968, -0.8591, -0.1871 ])
>>> torch.atanh(a)
tensor([ -1.7253,  0.3060, -1.2899, -0.1893 ])
```

Notes & Warnings

NOTE

The domain of the inverse hyperbolic tangent is $(-1, 1)$ and values outside this range will be mapped to NaN, except for the values 1 and -1 for which the output is mapped to $\pm\text{INF}$ respectively.

Detailed Documentation

Documentation

Returns a new tensor with the inverse hyperbolic tangent of the elements of input.

The domain of the inverse hyperbolic tangent is $(-1, 1)$ and values outside this range will be mapped to NaN, except for the values 1 and -1 for which the output is mapped to $\pm\text{INF}$ respectively.

input (Tensor) – the input tensor.

out (Tensor, optional) – the output tensor.

Returns a new tensor with the inverse hyperbolic tangent of the elements of input.

The domain of the inverse hyperbolic tangent is $(-1, 1)$ and values outside this range will be mapped to NaN, except for the values 1 and -1 for which the output is mapped to $\pm\text{INF}$ respectively.

input (Tensor) – the input tensor.

out (Tensor, optional) – the output tensor.